Yue Jiang

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Research Interests

Advanced Manufacturing Systems; Statistical Process Monitoring and Control; Fault Detection and Diagnosis; Active Learning; Reinforcement Learning; Large Language Model Agents

Education

- **Purdue University** *2022 – 2027 (Expected)*
  Ph.D. in Industrial Engineering, *Advisor: Dr. Ana María Estrada Gómez* West Lafayette, IN, USA
- **Shanghai Jiao Tong University** *2019 – 2022*
  M.S. in Industrial Engineering and Management Shanghai, China
- **Shanghai Jiao Tong University** *2015 – 2019*
  B.S. in Industrial Engineering and Management Shanghai, China

Academic Experience

- **Georgia Institute of Technology** *01/2025 – 04/2025*
  Visiting Researcher Atlanta, GA, USA
  H. Milton Stewart School of Industrial and Systems Engineering
  - Collaborated with Dr. Xiaochen Xian on active learning for online fault detection and diagnosis, supported by the Purdue TRACER Grant.

Honors and Awards

- **College of Engineering Conference Travel Grant**, Purdue University, 2026
- **IISE Future Faculty Fellow**, Institute of Industrial and Systems Engineers, 2025
- **TRAvel for CollaborativE Research (TRACER) Grant**, Purdue University, 2025
- **SPES+Q&P Student Paper Competition Winner**, American Statistical Association, 2024
- **Lee A. Chaden Fellowship in Industrial Engineering**, Purdue University, 2022–2023
- **Ross Fellowship**, Purdue University, 2022–2023

Publications

Published

[J1] **Jiang, Y.**, & Estrada Gómez, A. M. (2026). An Adaptive Sampling Strategy for Online Monitoring of Partially Observed Networks. *Technometrics*, *68*(2), 334–346. https://doi.org/10.1080/00401706.2025.2580634

[J2] **Jiang, Y.**, Xian, X., & Estrada Gómez, A. M. (2026). An Active Learning Strategy for Online Batch Fault Detection and Diagnosis. *INFORMS Journal on Data Science*.

[J3] **Jiang, Y.**, & Estrada Gómez, A. M. (2024). An Adaptive Sampling Strategy for Real-Time Anomaly Detection with Unmanned Sensing Vehicles. *Technometrics*, *66*(3), 438–454. https://doi.org/10.1080/00401706.2024.2322645
Winner, SPES+Q&P Student Paper Competition, American Statistical Association, 2024.

[J4] Li, N., Fang, T., **Jiang, Y.**, & Zhang, Z. H. (2025). Robust Valet Charging for Electric Vehicles with Uncertain Service Time. *Optimization and Engineering, 26*(3), 1895–1931. https://doi.org/10.1007/s11081-025-09960-5

[J5] Li, N., **Jiang, Y.**, & Zhang, Z. H. (2021). A Two-Stage Ambiguous Stochastic Program for Electric Vehicle Charging Station Location Problem with Valet Charging Service. *Transportation Research Part B: Methodological, 153*, 149–171. https://doi.org/10.1016/j.trb.2021.09.006

Working Papers

[1] **Jiang, Y.**, & Estrada Gómez, A. M. A Specification-Aware Residual Reinforcement Learning-Based Run-to-Run Control Framework for Semiconductor Manufacturing Processes. *Submitted to Journal of Quality Technology.*

[2] **Jiang, Y.**, Goñi, J., & Estrada Gómez, A. M. Online Structural Change-Point Detection of High-Dimensional Streaming Data via Sparse Spectral Graphical Models. *In preparation.*

[3] **Jiang, Y.**, Jurado Torres, E., Tay, L., & Estrada Gómez, A. M. A Personalized Model for Mental Health Forecasting Using Smartphone Sensing Data. *In preparation.*

Presentations

[1] **Jiang, Y.** (2026). Data-Driven Adaptive Modeling, Monitoring, and Control for Complex Processes in Advanced Manufacturing Systems. *Job Market Showcase Session, 2026 INFORMS Annual Meeting*, San Francisco, CA, USA.

[2] **Jiang, Y.**, & Estrada Gómez, A. M. (2025). An Adaptive Sampling Strategy for Real-Time Anomaly Detection with Unmanned Sensing Vehicles. *Invited Talk at the Technometrics Invited Session, 2025 INFORMS Annual Meeting*, Atlanta, GA, USA.

[3] **Jiang, Y.**, Xian, X., & Estrada Gómez, A. M. (2025). An Active Learning Strategy for Online Batch Fault Detection and Diagnosis. *Poster Presentation at the 2025 INFORMS Annual Meeting*, Atlanta, GA, USA.

[4] **Jiang, Y.**, Goñi, J., & Estrada Gómez, A. M. (2025). Online Structural Change-Point Detection of High-Dimensional Streaming Data via Sparse Spectral Graphical Models. *Oral Presentation at the 2025 IISE Annual Conference*, Atlanta, GA, USA.

[5] **Jiang, Y.**, & Estrada Gómez, A. M. (2024). An Adaptive Sampling Strategy for Online Monitoring of Partially Observed Networks. *Oral Presentation at the 2024 INFORMS Annual Meeting*, Seattle, WA, USA.

[6] **Jiang, Y.**, & Estrada Gómez, A. M. (2024). An Adaptive Sampling Strategy for Real-Time Anomaly Detection with Unmanned Sensing Vehicles. *Oral Presentation at the 2024 Joint Statistical Meetings*, Portland, OR, USA.

[7] **Jiang, Y.**, & Estrada Gómez, A. M. (2023). An Adaptive Sampling Strategy for Real-Time Anomaly Detection with Unmanned Sensing Vehicles. *Oral Presentation at the 2023 INFORMS Annual Meeting*, Phoenix, AZ, USA.

[8] **Jiang, Y.**, & Estrada Gómez, A. M. (2023). An Adaptive Sampling Strategy for Real-Time Anomaly Detection with Unmanned Sensing Vehicles. *Oral Presentation at the 2023 IISE Annual Conference*, New Orleans, LA, USA.

[9] **Jiang, Y.**, & Estrada Gómez, A. M. (2023). An Adaptive Sampling Strategy for Real-Time Anomaly Detection with Unmanned Sensing Vehicles. *Poster Presentation at the Statistics and Optimization in Data Science Workshop*, Purdue University, West Lafayette, IN, USA.

[10] **Jiang, Y.**, & Estrada Gómez, A. M. (2023). An Adaptive Sampling Strategy for Real-Time Anomaly Detection with Unmanned Sensing Vehicles. *Poster Presentation at the College of Agriculture and College of Engineering Joint Poster Session*, Purdue University, West Lafayette, IN, USA.

Teaching

- **Teaching Certificate** *2026*
  Certificate in Foundations of College Teaching, Purdue University
- **Teaching Assistant, Purdue University** *Spring 2026*
  IE 533 – Industrial Applications of Statistics
- **Teaching Assistant, Purdue University** *Fall 2025*
  IE 530 / STAT 513 – Quality Control
- **Guest Lecturer, Georgia Institute of Technology** *Spring 2025*
  ISyE 3039 – Methods for Quality Improvement
  - Lecture: Introduction to Multivariate Process Monitoring.
- **Guest Lecturer, Purdue University** *Fall 2023*
  IE 590 – Machine Learning and its Applications
  - Lecture: An Adaptive Sampling Strategy for Online Monitoring.

Mentorship and Outreach

Student Mentoring

Project: A Personalized Model for Mental Health Forecasting Using Smartphone Sensing Data, in collaboration with Dr. Louis Tay, Department of Psychological Sciences, Purdue University

- **Efrain Jurado Torres** *2026*
  Undergraduate Student in Applied Mathematics, Universidad EAFIT Medellín, Colombia
- **Charles Quan Liu** *2025*
  Undergraduate Student in Statistics, Purdue University West Lafayette, IN, USA

Outreach

- **Volunteer Mentor, Data Science in Bioengineering Workshop** *2024*
  Purdue University West Lafayette, IN, USA
- **Volunteer Mentor, Introduce a Girl to Engineering Day** *2023*
  Purdue University West Lafayette, IN, USA

Professional Service

- **Session Organizer, INFORMS Annual Meeting**
  - *Data-Driven Methods for Complex Data Modeling, Monitoring and Analysis in Advanced Manufacturing Systems*, 2026, San Francisco, CA, USA
  - *Data-Driven Strategies for Streaming Data Monitoring and Diagnosis*, 2025, Atlanta, GA, USA
- **Journal Reviewer**
  IISE Transactions; *IEEE Transactions on Automation Science and Engineering*; *IEEE Internet of Things Journal*; *IEEE Access*; *Journal of Applied Statistics*